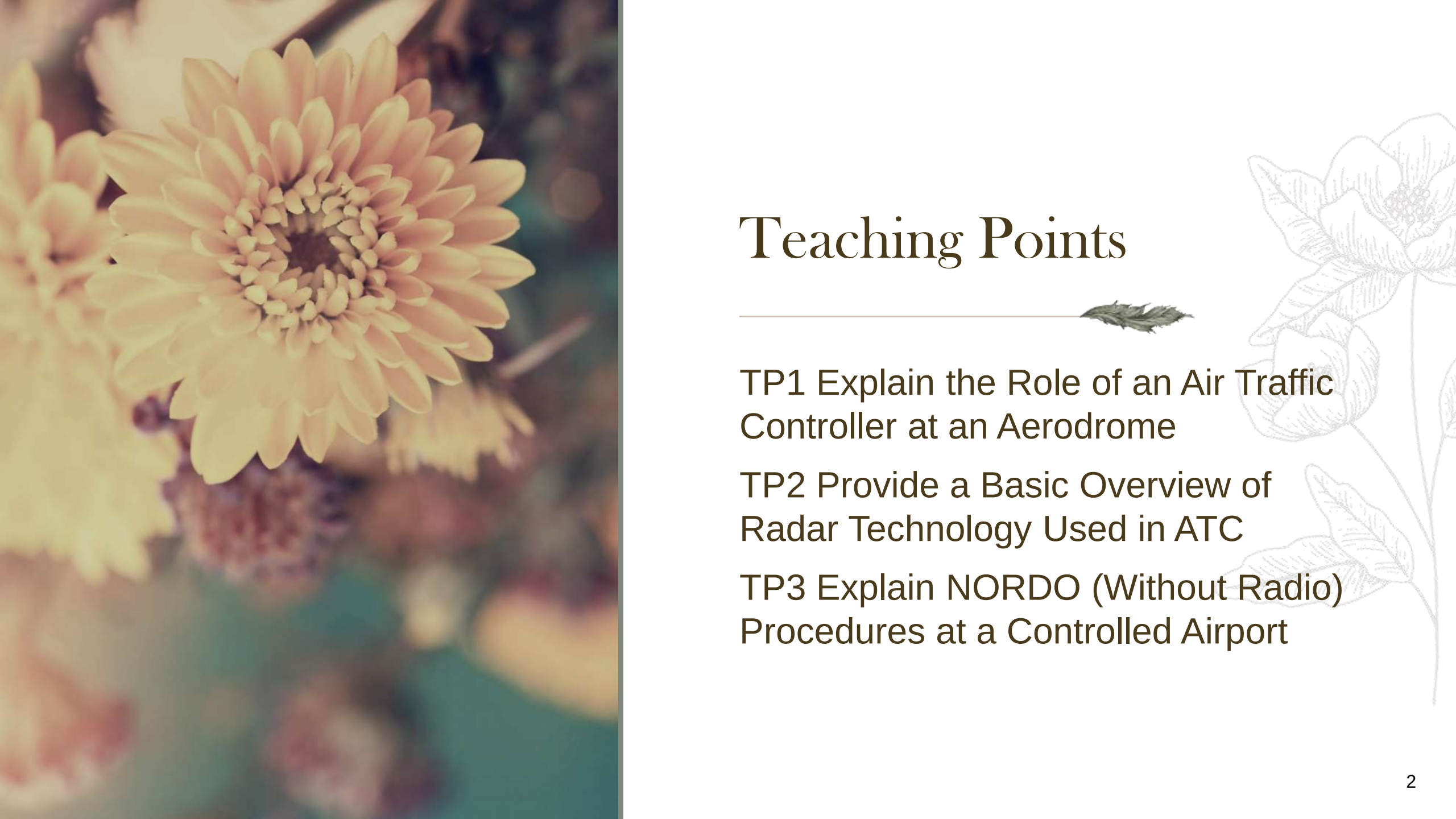


EO M260.01 –
EXPLAIN ASPECTS
OF AIR TRAFFIC
CONTROL (ATC)





Teaching Points

TP1 Explain the Role of an Air Traffic Controller at an Aerodrome

TP2 Provide a Basic Overview of Radar Technology Used in ATC

TP3 Explain NORDO (Without Radio) Procedures at a Controlled Airport



The power of communication

TP1 Explain the Role of an Air Traffic Controller at an Aerodrome



The ATC system is a **vast network of people and equipment** that ensures the safe operation of commercial and private aircraft.

The air traffic controller's' immediate concern is **safety**, but controllers must also direct planes efficiently to **minimize delays**. Their main responsibility is to **organize the flow of aircraft** and **prevent collisions** into and out of the aerodrome.

In addition, air traffic controllers keep pilots informed about **changes in weather conditions** such as wind shear, a sudden change in the velocity or direction of the wind that can cause the pilot to lose control of the aircraft

ATC Authorization



An ATC clearance is an authorization from an ATC unit for an **aircraft to proceed within controlled airspace under specific conditions**. Some air traffic controllers regulate traffic through designated airspaces; others regulate airport arrivals and departures.

TP2 Provide a Basic Overview of Radar Technology Used in ATC



RADAR

The name “RADAR” is an abbreviation for “radio detection and ranging”. To operate, radar requires a **highly directional radio transmitter/antenna** and a **scope/screen**, to display the information received by the antenna.

The use of radar in ATC greatly increases the utilization of the airspace and permits expansion of flight information services such as traffic and weather information and navigational assistance.

Used in:

- Fixing positions of airplanes in flight
- Detecting thunderstorm activity
- Approaching and landing guidance to airplanes

TP3 Explain NORDO (Without Radio) Procedures at a Controlled Airport



NORDO

Aircraft with **no radio** (NORDO) are **not permitted to operate** at most large controlled airports served by the scheduled air carriers.

Where they are permitted to operate (less busy controlled airports), they are **directed by visual signals**. A pilot must be alert to the light **signals from the tower letting you know what to do**.

Prior to initiating a NORDO flight, the pilot should contact the control tower to inform the controllers of their intentions and to secure a clearance for operation within the airspace. The tower will then be expecting the pilot and will be prepared to



Departing Aircraft

Arriving Aircraft

ATC Light Signals

GROUND	SIGNAL	AIR
Cleared for Takeoff		Cleared to Land
Cleared to Taxi		Return for Landing
STOP		Give Way Continue Circling
Taxi Clear of Runway		Airport Unsafe DO NOT LAND
Return to Starting Point on Airport		Not Applicable
Exercise EXTREME CAUTION		Exercise EXTREME CAUTION

